

Lovish Thukral *Software Developer & Engineer*

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Final-year Computer Science Engineering student specializing in the MERN stack, React Native, and native systems-level development (C++/Android-NDK/JNI). Built two production-grade native applications independently — a fully offline voice AI assistant and a vector-graphics file viewer with custom C++ parsing — demonstrating depth beyond typical web-stack tutorials. Comfortable across the full pipeline: REST APIs, database design, native module integration, and AI model orchestration. Seeking a Full Stack, React Native, or AI-integrated Developer role.

📁 PROJECTS

Vexa — Native Vector File Previewer, *React Native / C++ / Android NDK / libcdr / SQLite / TurboModule*

- Built a native document-processing pipeline to parse and preview CDR, AI, and PSD files on Android using custom JNI bindings with libcdr, enabling support for formats not natively available on the platform.
- Implemented a React Native New Architecture TurboModule (C++ codegen) to bridge native parsing logic with the JavaScript layer and handle proprietary binary file formats.
- Designed a SQLite-backed thumbnail caching system that stores generated previews in a dedicated cache, eliminating redundant parsing and improving large-file collection browsing performance

Sabrina AI — Fully Offline Voice AI Assistant, *Python / ONNX Runtime / faster-whisper / llama.cpp / ALSA / Local LLM*

- Built a fully offline, push-to-listen voice assistant using faster-whisper (int8-quantized), llama.cpp, and NeuTTS, eliminating cloud APIs and internet dependencies.
- Achieved 0.5-1s response latency after initial model load (~10-11s) on consumer hardware (16GB RAM, RTX 3050 6GB), demonstrating real-time edge AI performance.
- Developed an adaptive deployment system that analyzes device resources to automatically select the optimal LLM architecture and quantization tier, while processing all voice data locally for privacy.

Nextep AI – Intelligent Career Mentorship & Roadmap Platform,

React.js / Node.js / MongoDB / Express / Groq API / Tailwind CSS / Axios

- Engineered an AI agent platform using MongoDB and Groq API, featuring adaptive personality systems, contextual memory, and prompt orchestration to generate personalized interactions and dynamic learning roadmaps.

🏢 PROFESSIONAL EXPERIENCE

Full Stack Developer Trainee, *Vidya Corporation*

06/2025 – 07/2025 | Ludhiana

Completed a structured full-stack training program by building a MERN-based social media application featuring JWT authentication, posts, likes, comments, REST APIs, MongoDB schema design, and a responsive React frontend, while contributing to debugging and performance testing.

🎓 EDUCATION

Bachelor of Technology (B.Tech) - Computer Science Engineering,

07/2023 – Present

RIET, Phagwara | CGPA: 7.5+

🧠 SKILLS

Frontend: Next.js, React Native, Tailwind, Bootstrap, ShadenCN, GSAP, Framer

Backend & System: Node.js, REST APIs, MongoDB, SQLite, Redis, Android NDK, JNI, Authentication & Security, AWS

AI Systems: LLM Integration & Model Selection, Prompt Engg, Orchestration, ONNX Runtime, Quantization & Fine-Tuning

Languages: JavaScript, Python, TypeScript, C++

Dev Tools & Build Systems: Git, GitHub, Postman, Linux, Docker

🏆 ACHIEVEMENTS

Participant - Hack-n-Win Hackathon

Engineered an end-to-end career development platform featuring a voice-integrated mock interview system using React, Node.js, and MongoDB.

Open Source Contributor | GirlScript Summer of Code (GSSOC)

Contributed to multiple high-impact repositories by optimizing codebase performance and implementing new features across diverse tech stacks.